Google Cloud

Innovating with Google Cloud Artificial Intelligence

This study guide covers the materials from the third course of the six-course series for the Cloud Digital Leader certification.

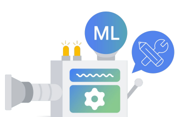
Certification exam focus

The Cloud Digital Leader certification exam is designed to assess knowledge of cloud literacy, common business use cases, and how cloud solutions and the capabilities of Google Cloud products support an enterprise.

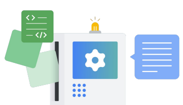
What is artificial intelligence (AI)?

AI is a broad field of computer science dedicated to building systems that can simulate human intelligence to perform tasks. Tasks include learning, reasoning, problem-solving, and decision-making.

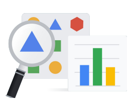
Fundamental AI concepts



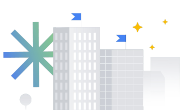
Machine learning (ML): A subset of AI with the ability to automatically learn and improve from experience without being explicitly programmed. ML uses algorithms to analyze data, identify patterns, and make predictions or decisions.



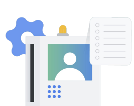
Generative AI (gen AI): A type of AI that can produce new content, including text, images, audio, and synthetic data. Common uses include chatbots, content generation, and document synthesis.



Data analytics: The science of examining raw data to draw conclusions. It involves various techniques and processes to clean, transform, and interpret data to uncover insights and support decision-making.



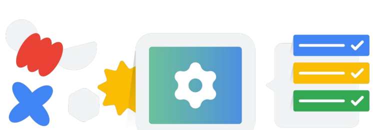
Business intelligence: A set of strategies, processes, apps, data, and technologies used to collect, analyze, present, and monitor business information. The primary goal is to improve business decisions by providing historical, current, and predictive views of business operations.



Agentic AI: A type of AI that is fundamentally reshaping the way people work. Individuals can use AI systems to interact with each other to complete complex, multistep workflows. Some use cases include workforce productivity, customer support, product innovation, operations, and research.

Responsible and explainable AI

Responsible AI refers to a governance framework covering fairness and accountability of AI systems.



Explainable AI is a specific subset of this framework that is focused on the internal logic and interpretability of model outputs. Responsible and explainable AI both aim to build trust and mitigate risk by ensuring technology is used ethically and safely.

The importance of high-quality data

The accuracy of ML predictions relies on large volumes of data. If data has errors, is not aligned to the problem, or is biased in some way, it is considered low quality. Data is evaluated against six dimensions to ensure that it is high quality:

- Completeness
- Uniqueness
- Timeliness
- Validity
- Accuracy
- Consistency

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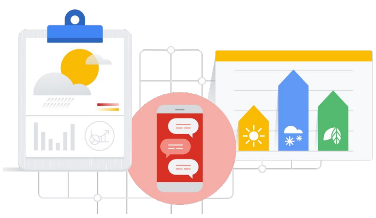
Business problems that AI can help solve

- Replacing or simplifying rule-based systems.
- Deriving business insights from large datasets (both structured and unstructured).
- Scaling business decisions.

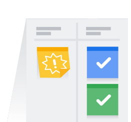


Key benefits of Google Cloud AI offerings

- Best infrastructure for AI
- AI-ready data cloud
- Sophisticated 1P models
- All-in-one AI developer platforms
- Prebuilt AI agents and apps

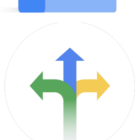
Consider these factors when selecting a Google Cloud AI solution for your business:

Implementation speed. How quickly do you need to get your model to production?

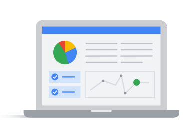


Potential for business differentiation. How unique is your model?

Technical expertise requirements. Does your staff need upskilling and training to implement an AI strategy?

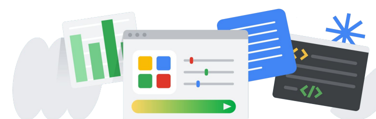


Choice and flexibility. Does the solution require customization, or will it use managed services?



Development effort. How complex is the problem, and how much data is available?

Gemini Enterprise Agent Platform: A comprehensive platform to build, scale, govern, and optimize agents.



Google Cloud pretrained APIs

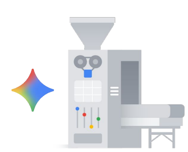
Gemini Enterprise Agent Platform API: Builds, orchestrates, and governs autonomous enterprise AI agents. Use case: Workflow automation.

Vision API: Extracts text, objects, and insights from visual imagery. Use case document OCR.

Cloud translation API: Dynamically translates text across thousands of language pairs. Use case: Website localization.

Speech-to-text API: Converts spoken audio into accurate written text. Use case: Call transcription.

Other Google Cloud AI solutions



Gemini: Google's advanced multimodal reasoning and generation model. Use case: Software development.

Agent Studio on Gemini Enterprise Agent Platform: Development workspace used to build autonomous, high-value business agents by grounding foundation models in proprietary data, visually designing operational logic, and integrating APIs to automate complex workflows.

AI Hypercomputer: Maximizes AI workload performance and efficiency by integrating optimized hardware like GPUs and TPUs with open software standards and flexible, cost-controlled consumption models. Use case: Supercomputing workloads.

BigQuery ML: Builds and executes ML models using standard SQL. Use case: Churn prediction.